MODULE 2

CENTRAL TENDENCY

OUTLINE

1. Central tendency: Mean, median, and mode
2. Statistical properties of the mean
3. Between-group application: Smoking cessation clinical trial
4. Within-subjects application: Age and Body Satisfaction
5. Study questions

OUTLINE

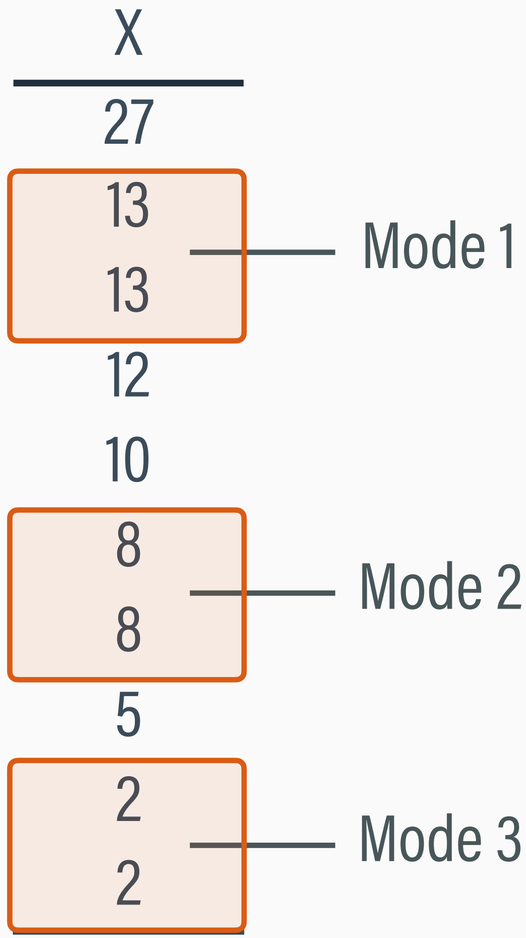
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CENTRAL TENDENCY

- Central tendency = the center of the distribution = a single value that is “typical” of the entire sample
- Three different definitions of center: mean (average score), median (middle score), and mode (most common score)
- We focus on the mean, as applications of the median and mode are uncommon in psychology research

THE MODE

- The mode is the most common score
- The only measure of central tendency for categorical variables
- Not useful for continuous variables because there can be multiple modes (e.g., 13, 8, and 2)



THE MEDIAN (50TH PERCENTILE)

- The median is the score that divides the distribution in half (half of the scores are above and below the median)

1. Rank scores then find the middle value
2. If even number, average the two middle scores

$$\text{median} = \frac{8 + 10}{2} = 9$$

X	
27	
13	
13	
12	
10	50th %tile
8	
8	
5	
2	
2	

THE MEAN (ARITHMETIC AVERAGE)

- The mean is the most common measure of central tendency for numeric variables in psychological research
- Among other things, psychology researchers use the mean to test hypotheses, compare groups, and examine trends over time
- The mean has desirable statistical properties that make it preferable (e.g., theorems that quantify the difference between a sample mean and the true population mean)

THE SAMPLE MEAN

- The arithmetic mean adds up scores and divides by the number of scores (the sample size, N)
- Conceptually, the mean distributes the total amount of the variable (the sum) equally across all sample members (N)

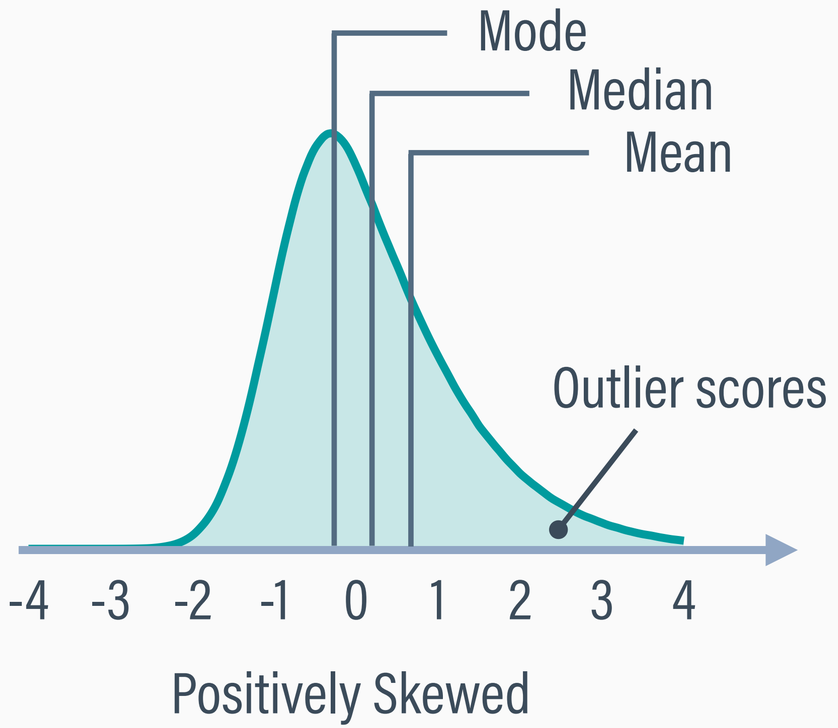
$$\bar{X} = \frac{\text{sum all scores}}{\text{number of scores}} = \frac{\sum X}{N}$$

$$= \frac{12 + 2 + 8 + 13 + 27 + 5 + 13 + 8 + 10 + 2}{10} = 10$$

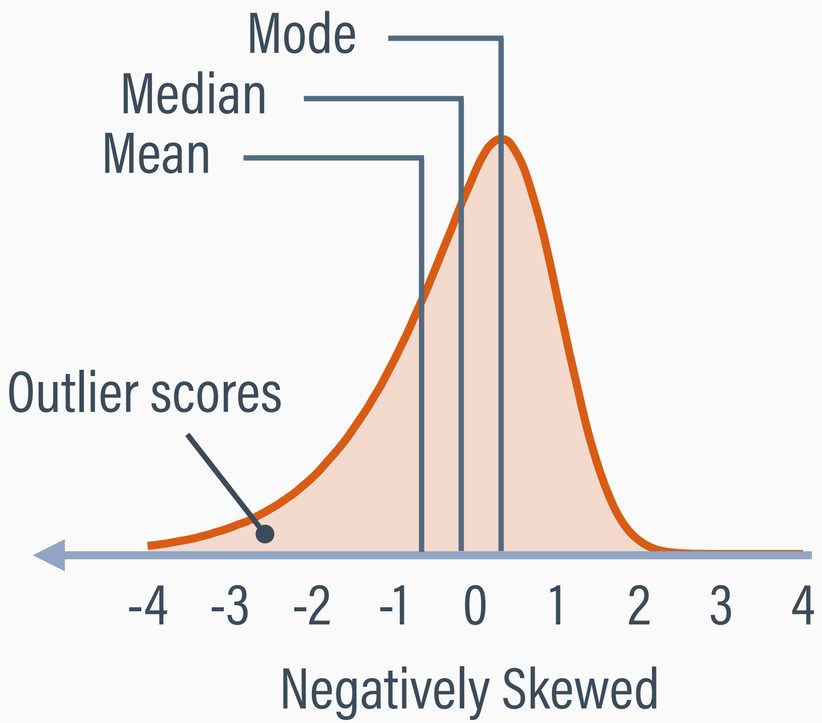
X
12
2
8
13
27
5
13
8
10
2
Sum = 100

CENTRAL TENDENCY IN SKEWED DISTRIBUTIONS

- Positively skewed distributions have extreme high scores in the right tail
- The high outlier scores contribute to the mean, pulling it toward the upper tail
- The mode is at the peak, the median cuts the distribution in half



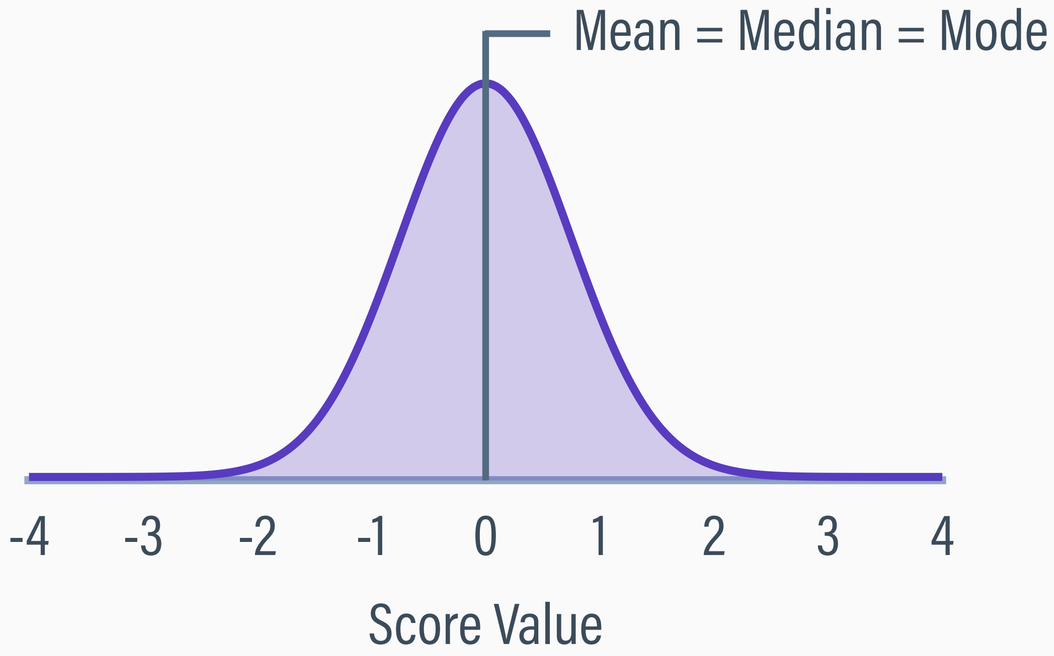
CENTRAL TENDENCY IN SKEWED DISTRIBUTIONS



- Negatively skewed distributions have extreme low scores in the left tail
- The low outlier scores contribute to the mean, pulling it toward the lower tail
- The mode is at the peak, the median cuts the distribution in half

CENTRAL TENDENCY IN SYMMETRIC DISTRIBUTIONS

- Extreme scores exist in both tails
- High and low outliers cancel out
- All three measures of central tendency are equal (or approximately so)



OUTLINE

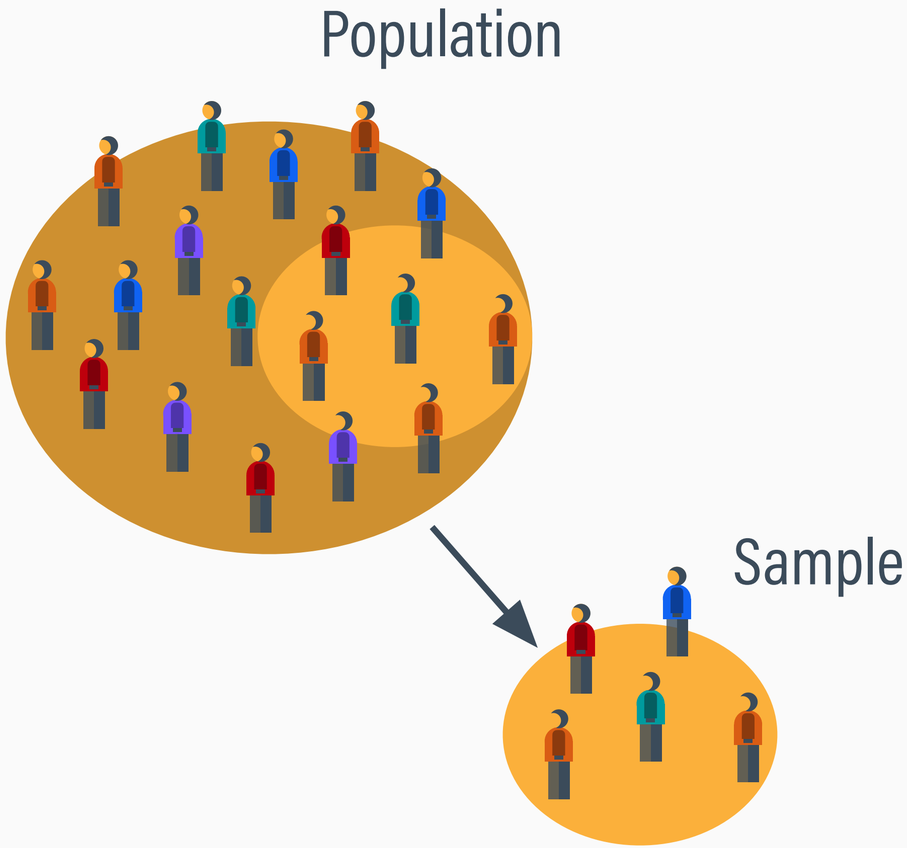
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IMPORTANT TERMINOLOGY

Participants	Statistic
Population = all possible participants who share an attribute of interest	Parameter = a hypothetical statistic computed using the full population The true value of statistics in population
Sample = the subset of people who participated in the study	Estimate = an observed statistic computed from the sample data

POPULATIONS VS. SAMPLES

- A population is the entire group of individuals that are of interest in a study
- Researchers almost exclusively work with a smaller subset called a sample
- The usual goal is use a sample statistic as a best guess about the population statistic



PARAMETERS VS. ESTIMATES

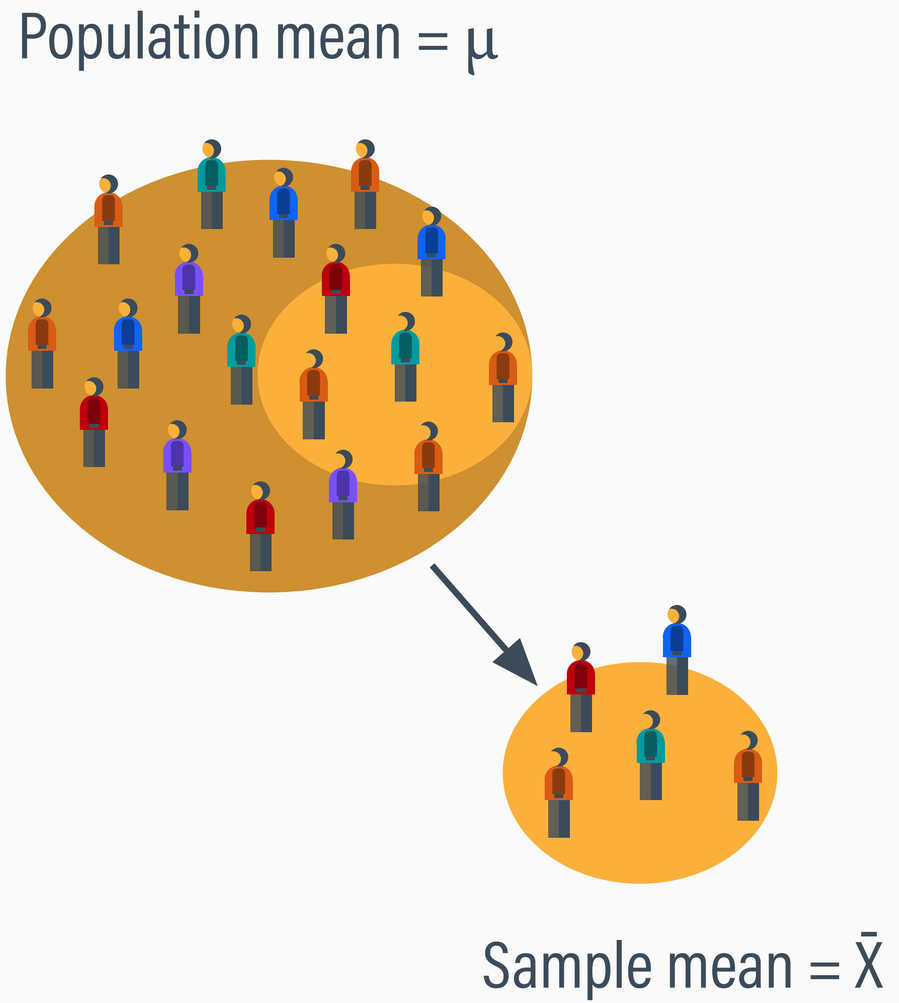
Greek letter is ‘parameter’?!

- The Greek letter μ denotes the true population-level mean (the parameter)

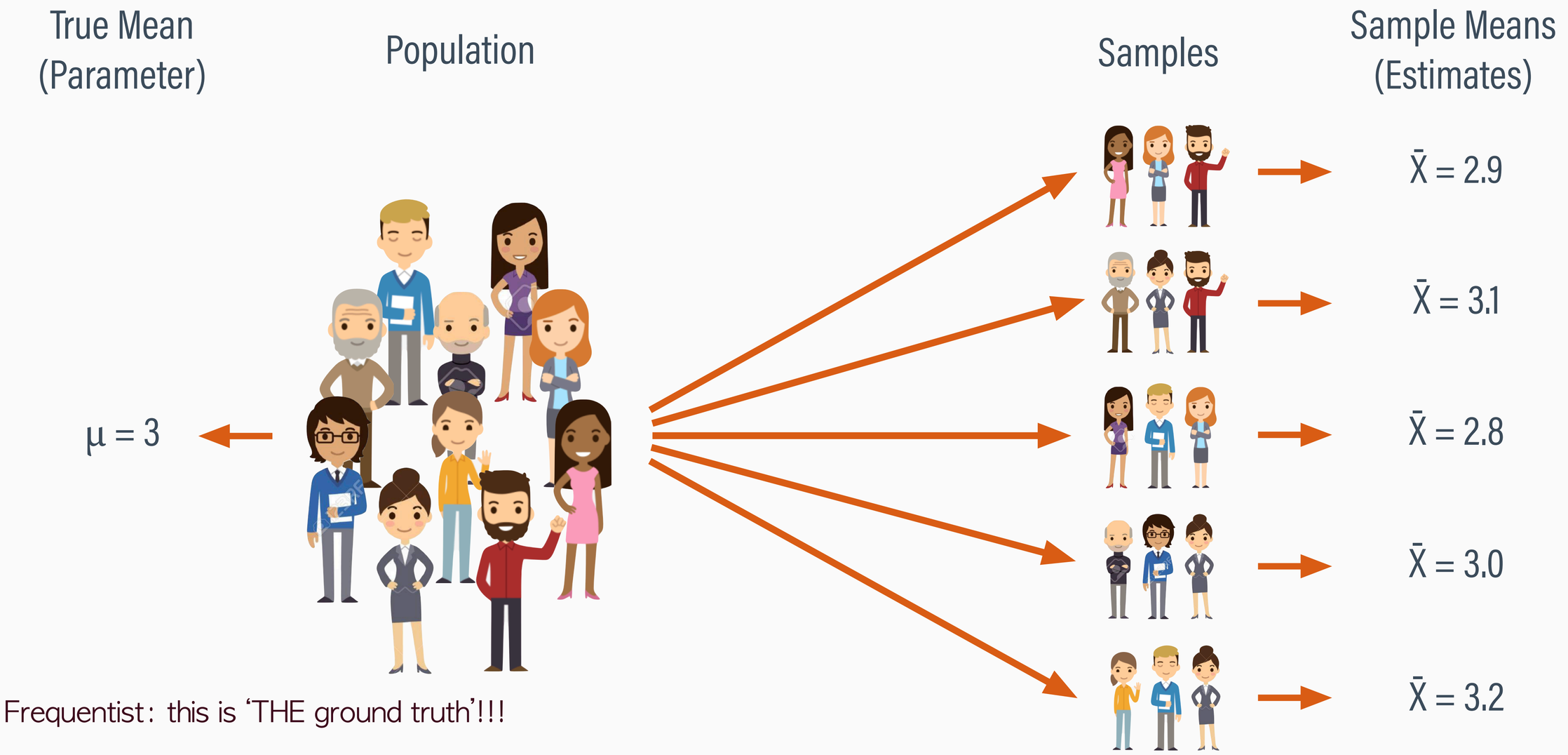
Roman letter is from ‘data’ ?!

- The Roman letter $\bar{X}$ references a sample mean (an estimate)

- Ideally, the sample mean will be close to the true mean, but it will always differ



SAMPLE MEANS DIFFER FROM THE TRUE MEAN



Frequentist: this is 'THE ground truth'!!!

UNBIASEDNESS OF SAMPLE MEAN

‘LOTS’ of samples of data 〉 Mean is an accurate snapshot of the mean of the true population.

- Each sample contains different individuals, so $\bar{X}$ will vary from sample to sample

- Some $\bar{X}$ will overestimate μ, others will underestimate it; some $\bar{X}$ will be very close to μ, others will be more distant

- The sample mean is an unbiased estimate of the true population mean: on average (across many samples), it doesn’t systematically miss in one direction

UNBIASED ESTIMATES

Estimates do not
systematically miss in one direction



Population

$\mu = 3$

2.7 — 2.8 — 2.9 — 3 — 3.1 — 3.2 — 3.3







$\bar{X} = 2.8$ $\bar{X} = 2.9$ $\bar{X} = 3.0$ $\bar{X} = 3.1$ $\bar{X} = 3.2$

Samples



In small groups of two or three, each of you should run the R code below, which draws a random sample of N = 100 scores from a population with a true mean of 3. Discuss how your sample means differ from the true mean of 3. Were they higher or lower? On average, about how much did they differ from the true mean?

```
mysample <- rnorm(n = 100, mean = 3, sd = 1)
mysample
hist(mysample)
cat("My sample mean = ", mean(mysample))
```

OUTLINE

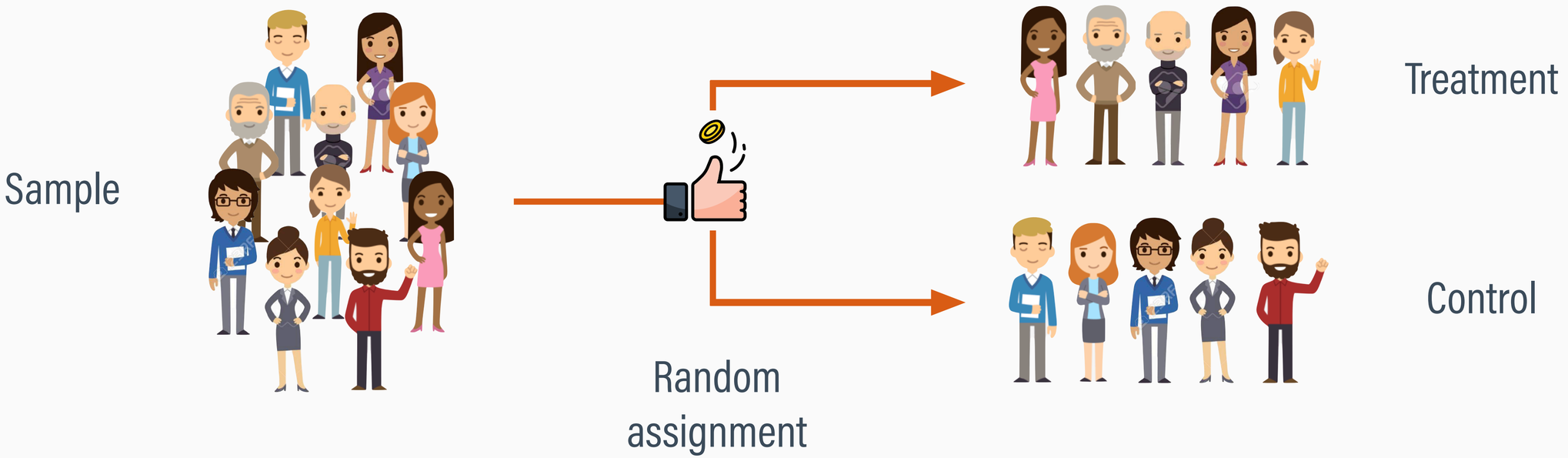
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BETWEEN-GROUP RESEARCH DESIGNS

- A between-group research design seeks to compare two or more groups of participants
- Unlike the within-group design, each condition is comprised of different participants
- The classic example is a randomized experiment with a treatment and control group, but groups can reflect any qualitative characteristic (e.g., sociodemographic)

EXPERIMENTAL APPLICATION

- Participants are randomly assigned to either a treatment or a control condition



NON-EXPERIMENTAL APPLICATION

- Participants divide into subgroups based on a shared qualitative characteristic

Sample



Pre-existing groups



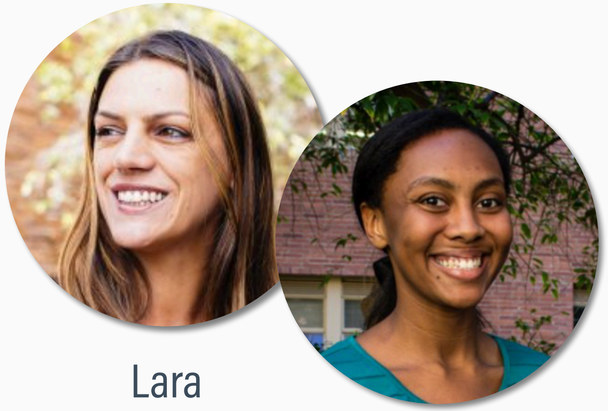
Females



Males

SMOKING AND DRINKING CESSATION TRIAL

Pharmacological treatments that can concomitantly address cigarette smoking and heavy drinking stand to improve health care delivery for these highly prevalent co-occurring conditions. This superiority trial compared the combination of varenicline and naltrexone against varenicline alone for smoking cessation and drinking reduction among heavy-drinking smokers.



Lara Ray

ReJoyce Green

Ray, L.A., Green, R., Enders, C., et al. (2021). Efficacy of combining varenicline and naltrexone for smoking cessation and drinking reduction: A randomized clinical trial. *American Journal of Psychiatry*, *178*, 818–828.

KEY VARIABLES



Breath (alveolar) carbon monoxide

A measure of carbon monoxide in the lungs. Breath carbon monoxide is a biomarker of smoking behavior common in clinical trials. Higher scores reflect more frequent smoking.

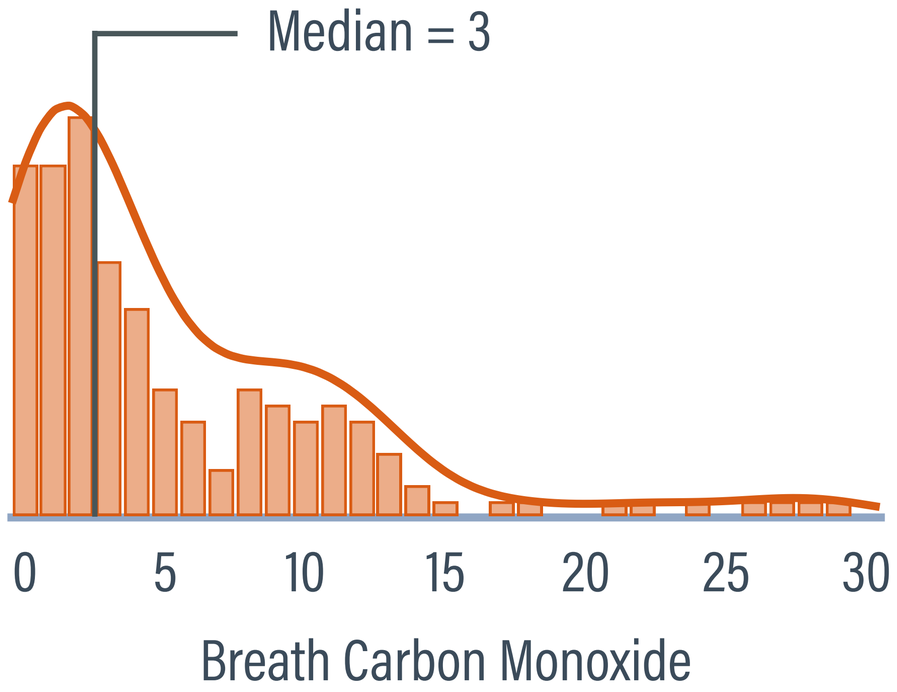


Medication arm

Participants were randomly assigned to receive one of two meds: varenicline plus naltrexone or varenicline plus placebo pills

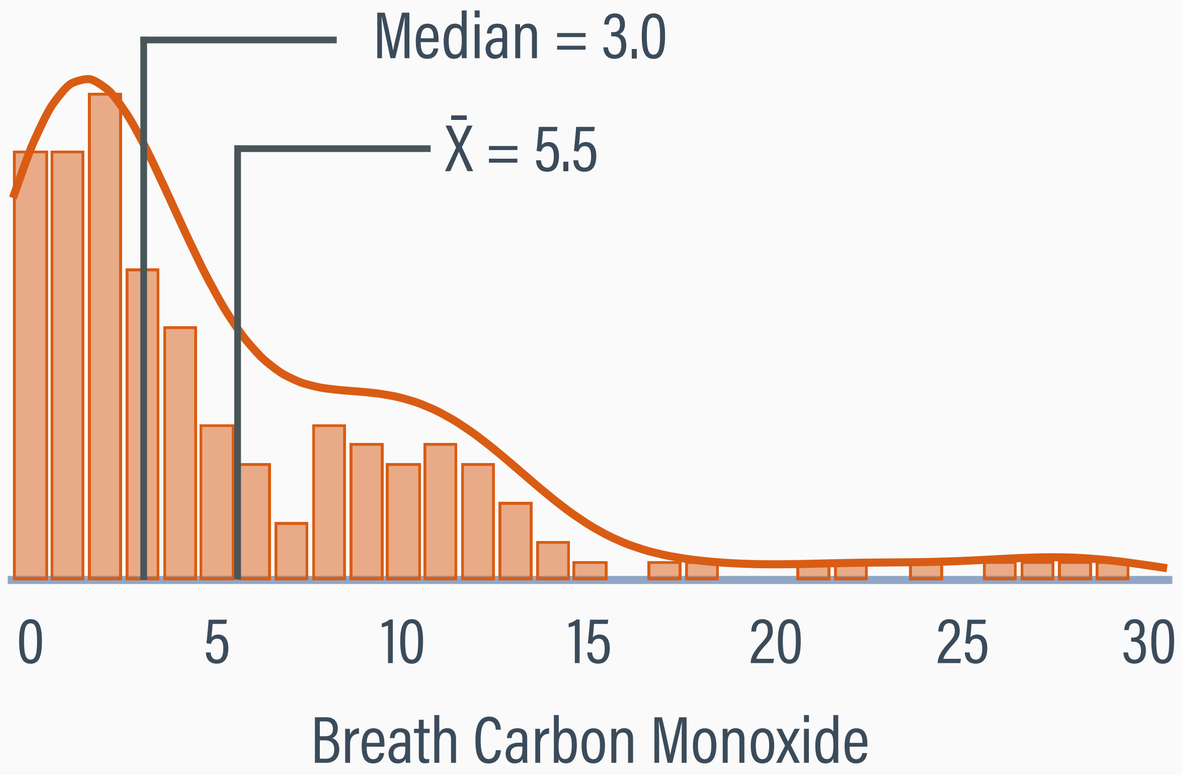


The median breath CO reading at the eight-week follow-up is 3.0. In small groups of two or three, discuss whether the sample mean is higher, lower, or about the same as than the median. Provide a rationale.



MEAN VS. MEDIAN

- High CO values in the upper tail pull the mean above the median



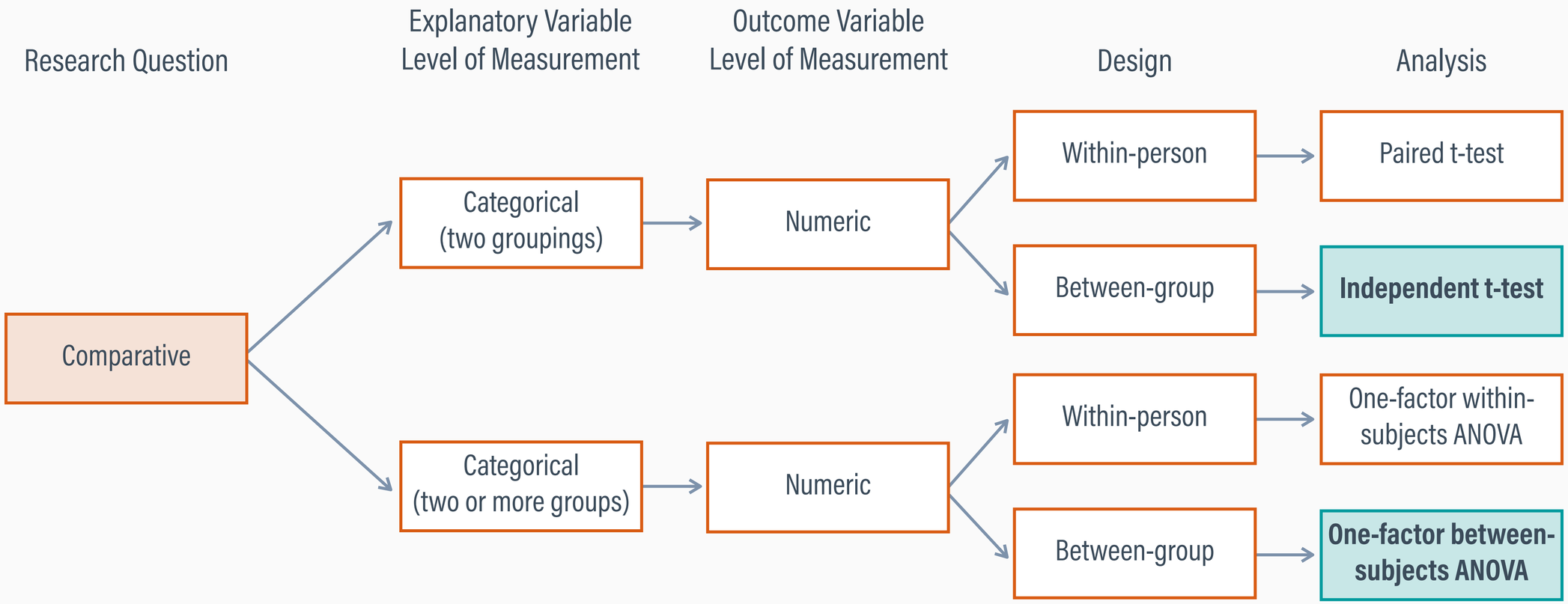
R OUTPUT

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
Participant	1	165	83.00	47.78	83	83.00	60.79	1	165	164	0.00	-1.22	3.72
Condition*	2	165	1.50	0.50	2	1.50	0.00	1	2	1	-0.01	-2.01	0.04
Gender*	3	165	1.61	0.49	2	1.63	0.00	1	2	1	-0.43	-1.83	0.04
COWeek0	4	165	10.55	6.86	9	9.82	5.93	0	46	46	1.41	3.60	0.53
COWeek4	5	165	5.46	5.10	4	4.69	4.45	0	24	24	1.24	0.96	0.40
COWeek8	6	165	5.53	5.96	3	4.55	2.97	0	29	29	1.75	3.35	0.46
QuitCigsWeek4*	7	165	1.62	0.49	2	1.65	0.00	1	2	1	-0.48	-1.78	0.04
QuitCigsWeek8*	8	165	1.64	0.48	2	1.68	0.00	1	2	1	-0.59	-1.66	0.04
DrinksWeek0	9	165	6.40	4.42	6	5.77	2.97	1	35	34	2.44	10.46	0.34
DrinksWeek4	10	165	3.59	2.98	3	3.30	2.97	0	13	13	0.80	0.30	0.23
DrinksWeek8	11	165	3.23	2.68	3	2.97	2.97	0	13	13	0.89	0.67	0.21
CigsWeek0	12	165	14.22	8.22	12	13.05	5.93	3	51	48	1.59	3.20	0.64
CigsWeek4	13	165	4.18	5.70	2	3.08	2.97	0	41	41	3.01	13.13	0.44
CigsWeek8	14	165	3.16	4.75	2	2.25	2.97	0	35	35	3.44	16.81	0.37

COMPARATIVE RESEARCH QUESTIONS

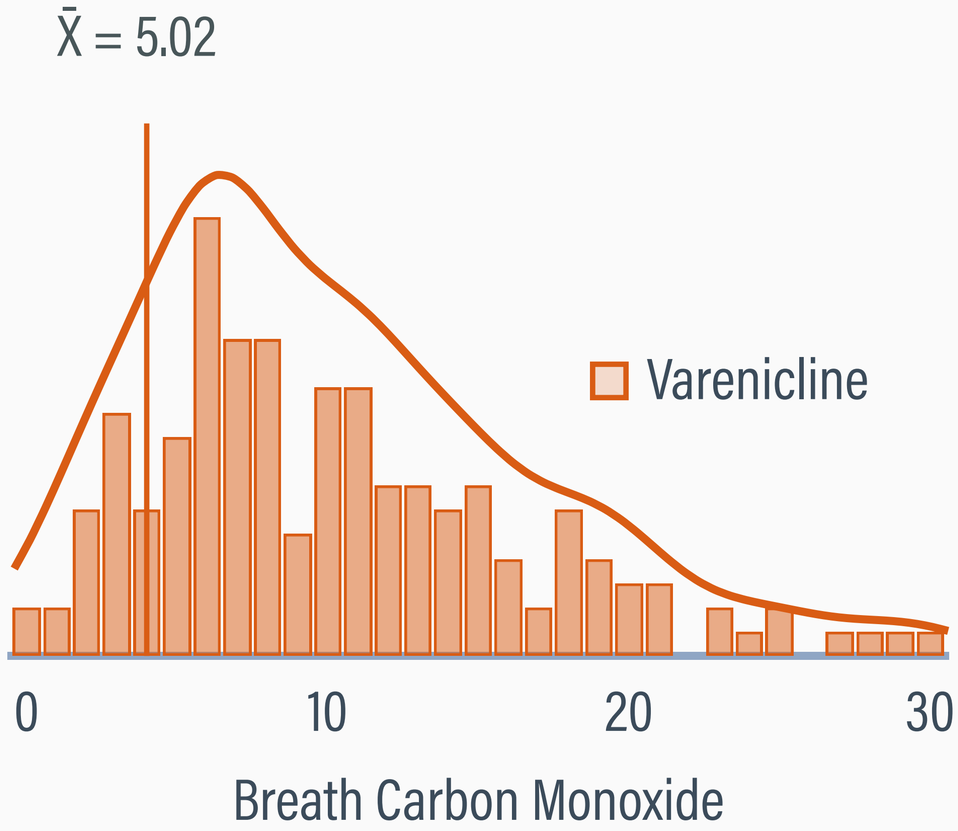
- Comparative research questions ask whether two or more groups (or occasions) differ from one another
- Question: Do participants in the two treatment groups differ in their smoking levels?
- We can answer this question by comparing the group means

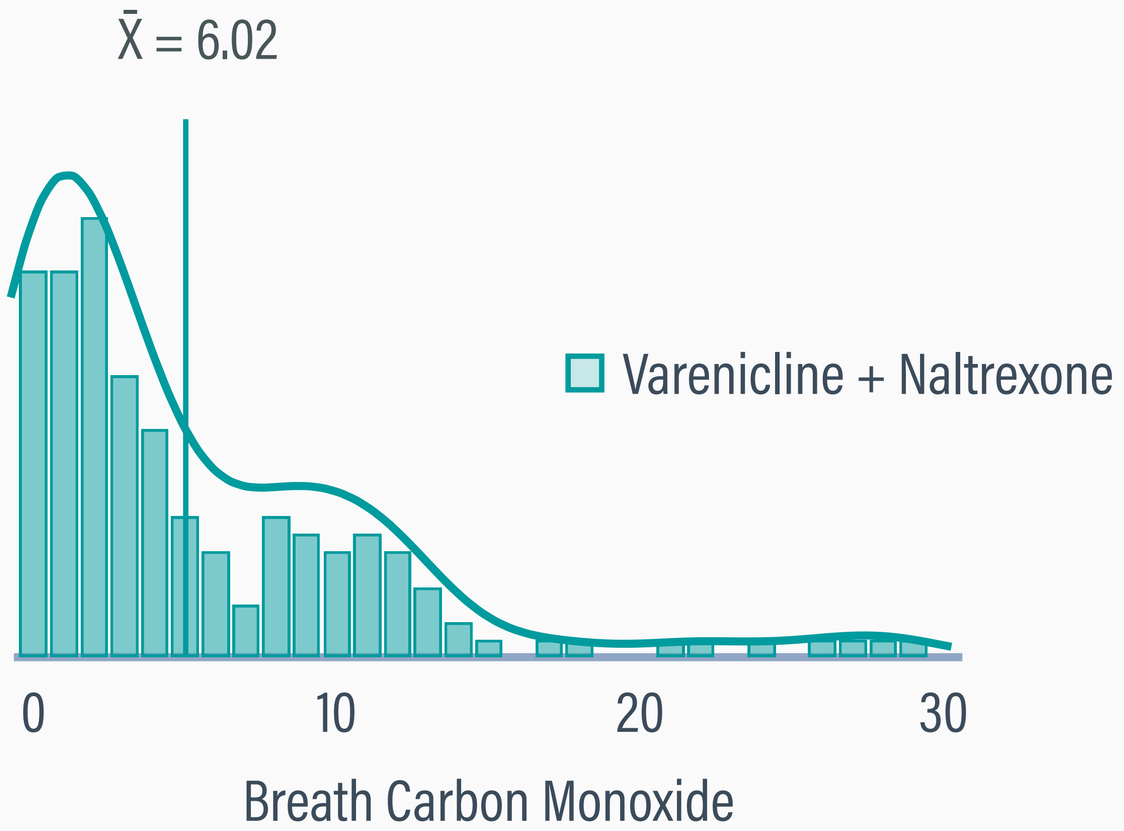
STATISTICAL ORG CHART



One-factor =
One independent categorical varaible

GROUP MEANS


X̄ = 5.02
Varenicline
0
10
20
30
Breath Carbon Monoxide


X̄ = 6.02
Varenicline + Naltrexone
0
10
20
30
Breath Carbon Monoxide

R OUTPUT

Cohen's d effect size mentioned here

```
 Descriptive statistics by group
Condition: Varenicline
        vars  n mean   sd median trimmed  mad min max range skew kurtosis   se
COWeek8    1 82 5.02 4.88    3.5    4.33 3.71   0  26    26 1.57     3.09 0.54
------------------------------------------------------------------------------
Condition: Varenicline + Naltrexone
        vars  n mean   sd median trimmed  mad min max range skew kurtosis   se
COWeek8    1 83 6.02 6.86      3    4.81 4.45   0  29    29 1.62     2.31 0.75
```



One of the main goals of the research study is to determine whether the treatment groups differ. In small groups of two or three, discuss the size of the mean difference. How would you gauge the magnitude of the medication difference? Do the groups exhibit a meaningful difference? This is a subjective evaluation, so provide a rationale.

OUTLINE

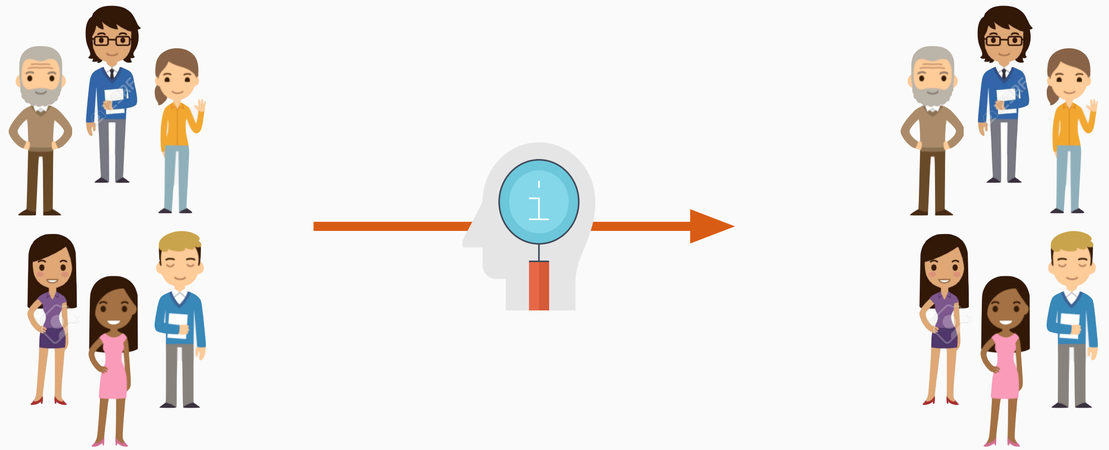
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WITHIN-SUBJECTS RESEARCH DESIGNS

- A within-subjects research design produces two or more scores from the same participant
- A classic within-group design collects two or more repeated measurements from the same group of individuals (e.g., a pretest measure followed by a posttest)
- The scores can also come from matched pairs of scores

INTERVENTION APPLICATION

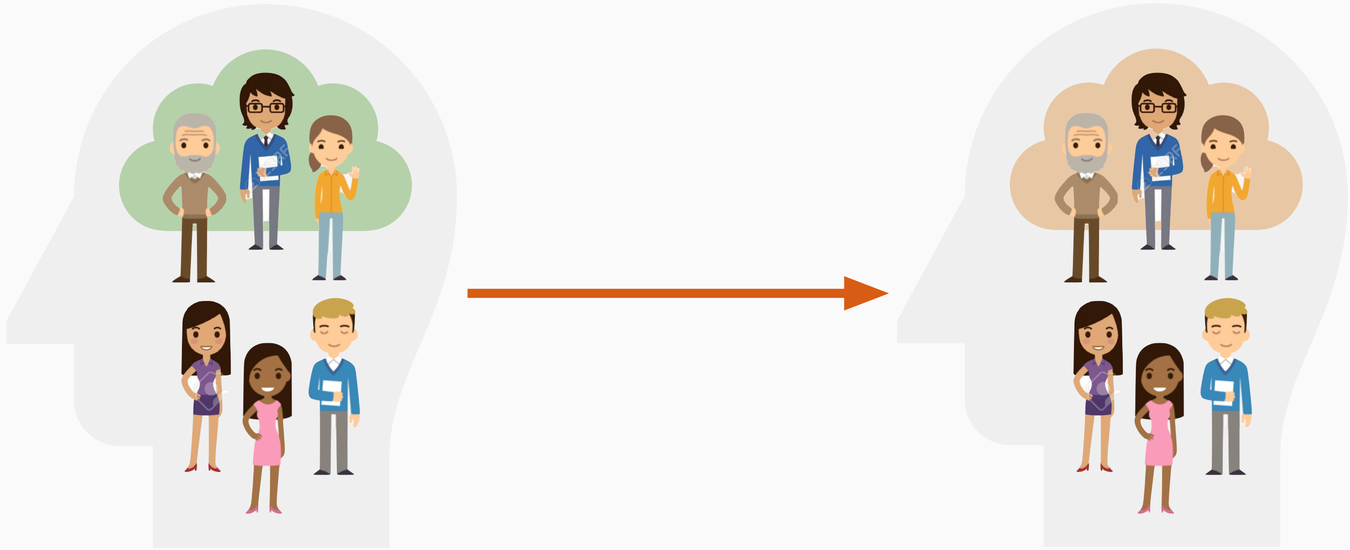
- Two scores are obtained from the same group of people before and after an intervention



Baseline Intervention Follow-up

EXPERIMENTAL APPLICATION

- The same group of people are exposed to two different experimental conditions

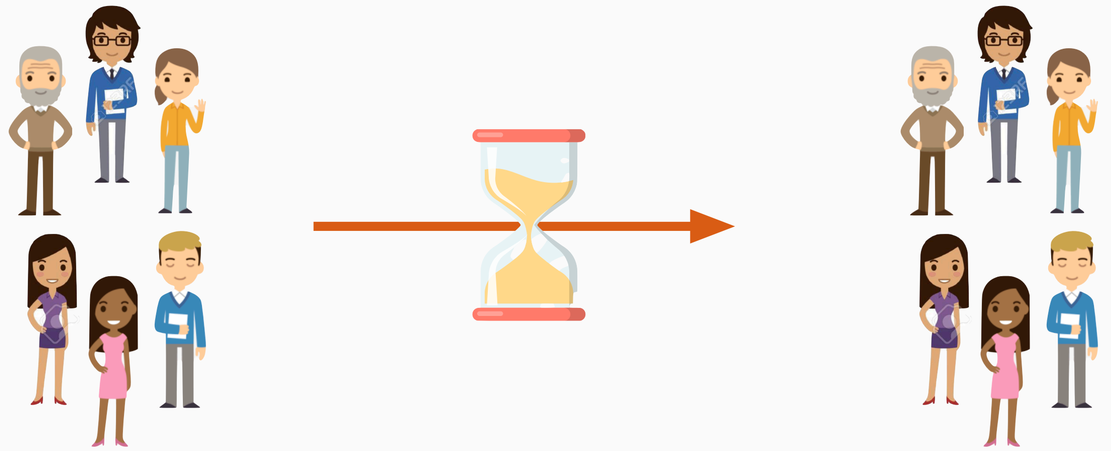


Stimuli 1

Stimuli 2

DEVELOPMENTAL APPLICATION

- The same group of people are followed over time, with the goal of examining change or development

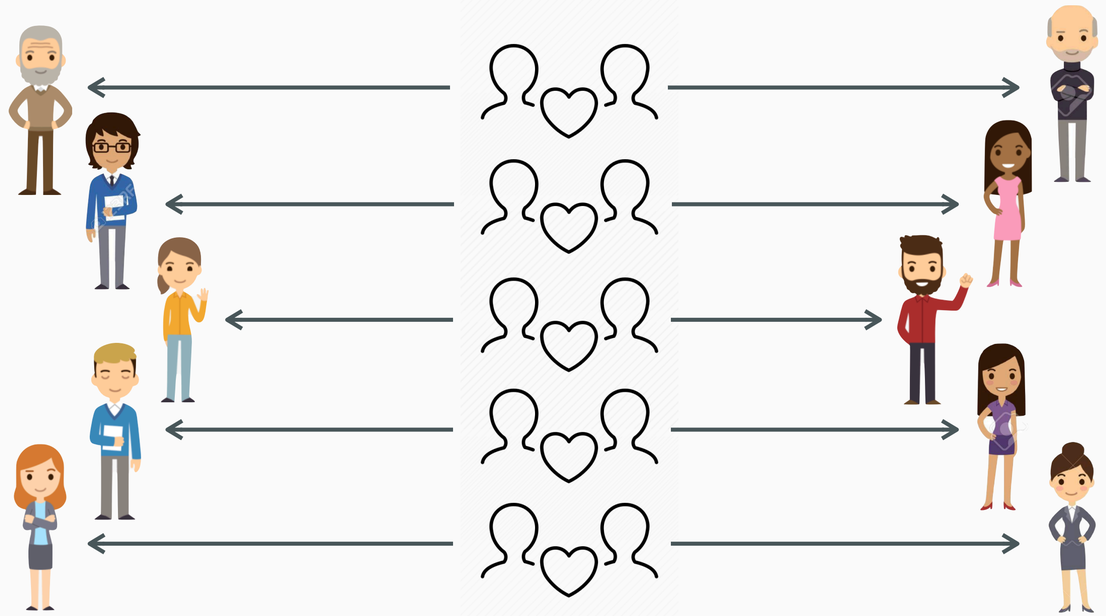


Occasion 1

Occasion 2

DYADIC APPLICATION

- Pairs of individuals forming naturally-occurring dyads (e.g., romantic partners, siblings) with linked scores



KEY VARIABLES



Breath (alveolar) carbon monoxide

A measure of carbon monoxide in the lungs. Breath carbon monoxide is a biomarker of smoking behavior common in clinical trials. Higher scores reflect more frequent smoking.



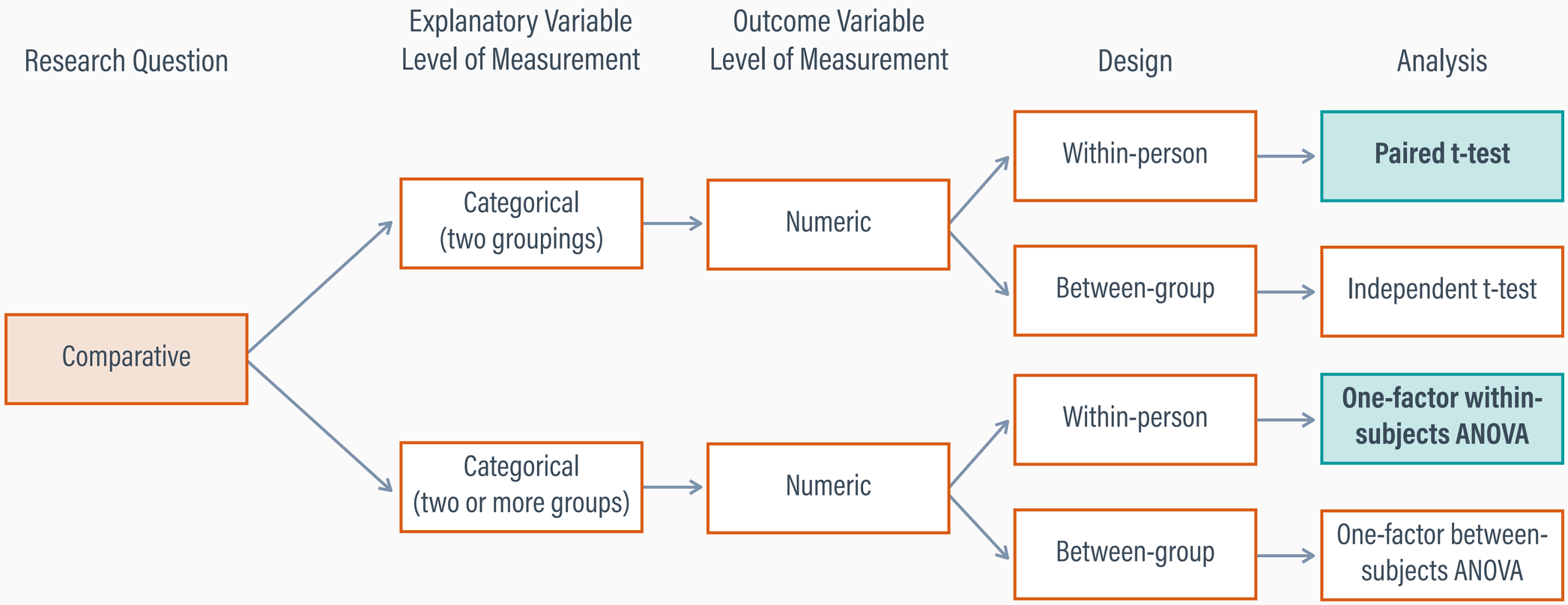
Measurement occasion

Three repeated measurements collected at baseline (before treatment), at the 4-week follow-up, and at the 8-week follow-up.

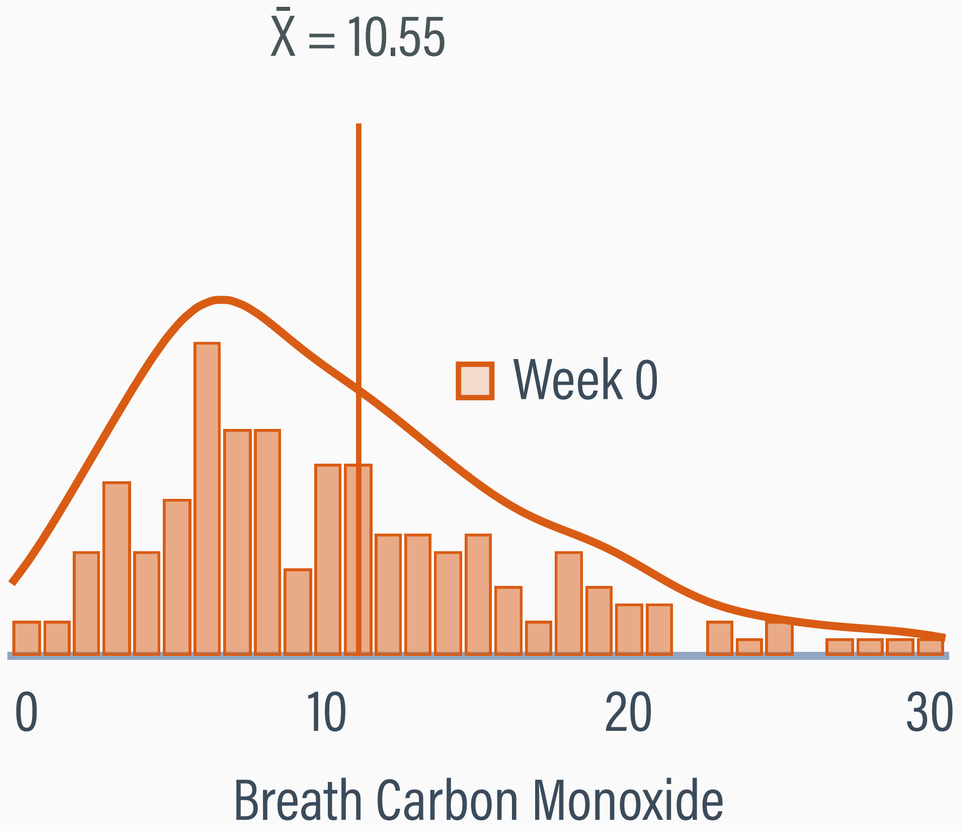
COMPARATIVE RESEARCH QUESTIONS

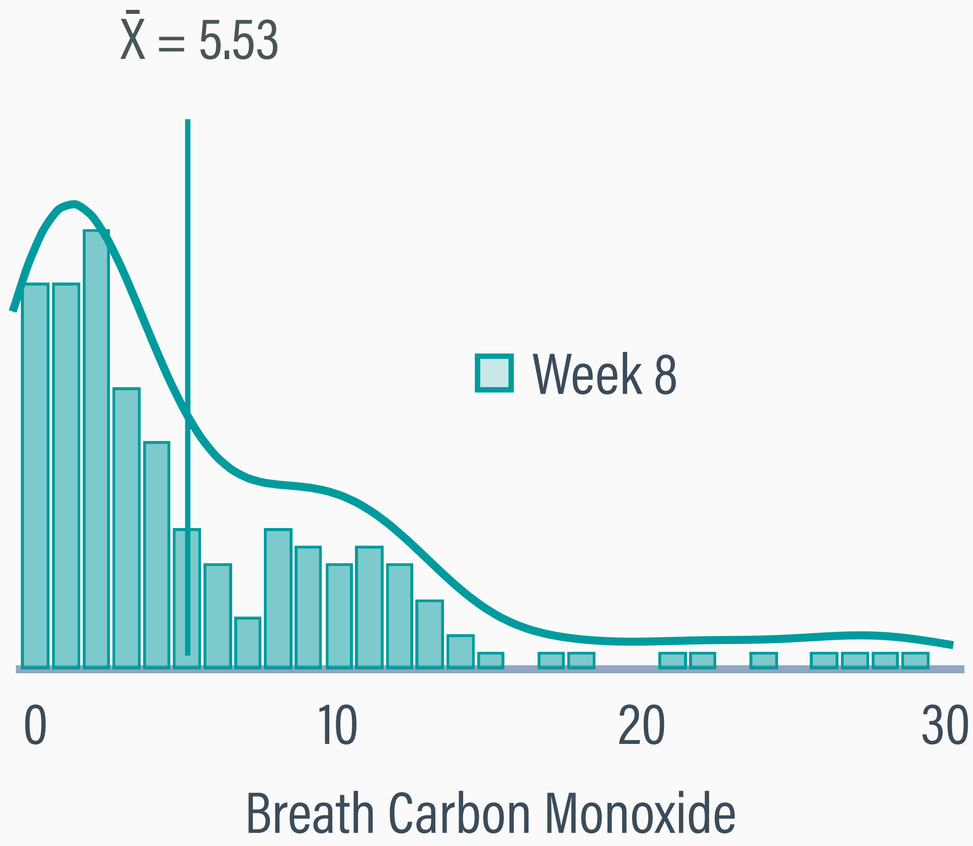
- Comparative research questions ask whether two or more groups (or occasions) differ from one another
- Question: Does breath CO change between the baseline assessment and the 8-week follow-up?
- We can answer this question by comparing the means

STATISTICAL ORG CHART



OCCASION-SPECIFIC MEANS





R OUTPUT

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
Participant	1	165	83.00	47.78	83	83.00	60.79	1	165	164	0.00	-1.22	3.72
Condition*	2	165	1.50	0.50	2	1.50	0.00	1	2	1	-0.01	-2.01	0.04
Male*	3	165	1.61	0.49	2	1.63	0.00	1	2	1	-0.43	-1.83	0.04
COWeek0	4	165	10.55	6.86	9	9.82	5.93	0	46	46	1.41	3.60	0.53
COWeek4	5	165	5.46	5.10	4	4.69	4.45	0	24	24	1.24	0.96	0.40
COWeek8	6	165	5.53	5.96	3	4.55	2.97	0	29	29	1.75	3.35	0.46
QuitCigsWeek4*	7	165	1.62	0.49	2	1.65	0.00	1	2	1	-0.48	-1.78	0.04
QuitCigsWeek8*	8	165	1.64	0.48	2	1.68	0.00	1	2	1	-0.59	-1.66	0.04
DrinksWeek0	9	165	6.40	4.42	6	5.77	2.97	1	35	34	2.44	10.46	0.34
DrinksWeek4	10	165	3.59	2.98	3	3.30	2.97	0	13	13	0.80	0.30	0.23
DrinksWeek8	11	165	3.23	2.68	3	2.97	2.97	0	13	13	0.89	0.67	0.21
CigsWeek0	12	165	14.22	8.22	12	13.05	5.93	3	51	48	1.59	3.20	0.64
CigsWeek4	13	165	4.18	5.70	2	3.08	2.97	0	41	41	3.01	13.13	0.44
CigsWeek8	14	165	3.16	4.75	2	2.25	2.97	0	35	35	3.44	16.81	0.37



One of the main goals of the research study is to determine whether cigarette smoking changes over time. In small groups of two or three, discuss the size of the mean difference. How would you gauge the magnitude of the mean difference? Do the occasion-specific means exhibit a meaningful difference? This is a subjective evaluation, so provide a rationale.

OUTLINE

1. Central tendency: Mean, median, and mode
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5. Study questions

STUDY QUESTIONS

- The study questions continue using the clinical trial research scenario introduced in the lecture, including new variables
- The questions illustrate the kinds of statistical reasoning the exams will require, including interpreting evidence, evaluating conclusions, and connecting graphical and numerical results
- Midterm exam questions will require you to apply these same reasoning skills to new research scenarios

R OUTPUT

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
Participant	1	165	83.00	47.78	83	83.00	60.79	1	165	164	0.00	-1.22	3.72
Condition*	2	165	1.50	0.50	2	1.50	0.00	1	2	1	-0.01	-2.01	0.04
Male*	3	165	1.61	0.49	2	1.63	0.00	1	2	1	-0.43	-1.83	0.04
COWeek0	4	165	10.55	6.86	9	9.82	5.93	0	46	46	1.41	3.60	0.53
COWeek4	5	165	5.46	5.10	4	4.69	4.45	0	24	24	1.24	0.96	0.40
COWeek8	6	165	5.53	5.96	3	4.55	2.97	0	29	29	1.75	3.35	0.46
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QuitCigsWeek8*	8	165	1.64	0.48	2	1.68	0.00	1	2	1	-0.59	-1.66	0.04
DrinksWeek0	9	165	6.40	4.42	6	5.77	2.97	1	35	34	2.44	10.46	0.34
DrinksWeek4	10	165	3.59	2.98	3	3.30	2.97	0	13	13	0.80	0.30	0.23
DrinksWeek8	11	165	3.23	2.68	3	2.97	2.97	0	13	13	0.89	0.67	0.21
CigsWeek0	12	165	14.22	8.22	12	13.05	5.93	3	51	48	1.59	3.20	0.64
CigsWeek4	13	165	4.18	5.70	2	3.08	2.97	0	41	41	3.01	13.13	0.44
CigsWeek8	14	165	3.16	4.75	2	2.25	2.97	0	35	35	3.44	16.81	0.37

STUDY QUESTIONS (1)

1. The maximum value of CigsWeek0 is 51. Suppose this score were changed from 51 to 30. Consider the mean, median, and mode. Which numeric values would change and why?

2. After the Male variable is converted to a factor, the R output reports a mean of 1.61 and a median of 2, based on internal values of 1 = female and 2 = male. Are the mean and median meaningful measures of central tendency for this variable? Explain what, if anything, each statistic reveals about the sample.

STUDY QUESTIONS (2)

3. This study included both between-subjects and within-subjects research design components. Describe how each component is represented in the study.

4. Suppose the number of drinks consumed during the previous week were recoded into an ordinal classification variable: 0 = nondrinker, 1 = nondaily drinker, and 2 = daily drinker. Could the mean, median, and mode be used to summarize this variable? For each statistic, explain why or why not.

STUDY QUESTIONS (3)

5. For CigsWeek0, the mean is 14.22 cigarettes and the median is 12 cigarettes. Both statistics describe a typical score, yet they differ by more than two cigarettes. What feature of the distribution explains this difference? Which statistic do you consider more representative of a typical participant's smoking level, and why?

6. Sketch a histogram or kernel density plot that depicts the location of each statistic in the distribution. Where is the mode relative to the mean and median?

STUDY QUESTIONS (4)

7. The study included a longitudinal component in which the number of cigarettes smoked was measured at baseline and again at the 8-week follow-up. A major goal of the study is to determine whether the medications improve smoking. State a hypothesis: What pattern of means would you expect across the two measurement occasions? Briefly explain your prediction.

8. Is your hypothesis consistent or inconsistent with the CigsWeek0 and CigsWeek8 means? Support your conclusion by comparing the baseline and 8-week means, describing the direction and magnitude of the change.

R OUTPUT

```
 Descriptive statistics by group
Condition: Varenicline
           vars  n mean   sd median trimmed  mad min max range skew kurtosis   se
CigsWeek8     1 82 3.43 5.08      2    2.44 2.97   0  35    35  3.5     16.8 0.56
------------------------------------------------------------------------------
Condition: Varenicline + Naltrexone
           vars  n mean   sd median trimmed  mad min max range skew kurtosis   se
CigsWeek8     1 83 2.89 4.42      1    2.06 1.48   0  30    30 3.18    14.93 0.49
```

STUDY QUESTIONS (5)

9. The mean number of cigarettes smoked at the 8-week follow-up was 3.43 in the varenicline group and 2.89 in the varenicline-plus-naltrexone group. Later in the quarter, we will learn about the null hypothesis that the two medication conditions have identical population means. Subjectively, explain whether you think the observed difference between the sample means is consistent or inconsistent with this null hypothesis.

10. Suppose that, in the full population of smokers, the two medication conditions produce the same mean number of cigarettes smoked at the 8-week follow-up. Explain why a sample of 165 participants drawn from this null population could nevertheless produce group means that are not identical.